LESSON PLAN

EWIS (21)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	10	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to sonar system
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to sonar system. Purpose and technical details of active and passive sonar system.
5. Specific Objectives	21.1.1.1 Trainee will be able to understand the basic theory of communication.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of sonar theory diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a sonar system?
(b) Define the basic theory of communication in underwater contexts.
(c) List two applications of sonar in naval operations.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Basic theory of communication • Sound propagation in water • Active sonar operating principle • Passive sonar operating principle • Comparison of active and passive systems	L & Ds	PPT, IB	15 min
(b)	• Technical details of active sonar • Technical details of passive sonar • Signal processing basics • Transducer types and functions • Echo location and target detection	L & Ds	PPT, PDF	20 min
(c)	• Limitations of active sonar • Limitations of passive sonar • Environmental factors affecting sonar • Operational constraints • Future trends in sonar technology	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between active and passive sonar detection methods.
(b) Describe how sound waves are utilized for underwater communication.
(c) Identify one key limitation of active sonar systems.

12. Home Assignment and Follow Up	Prepare a brief note on the transition from LP 05 to LP 06, focusing on how sonar data feeds into sonobuoy systems.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Introduction to sonobuoy system. Purpose and technical details of active and passive sonobuoy.	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min